# Assignment 1 - Git & GitHub Fundamentals

Report

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## 1. Repository Setup

I initialized `git-session1-lurantys` with `git init`. The `.gitignore` excludes `local.env` and regular `.log` files; an exception keeps `important.log` trackable.

## 2. README & Commit History

My README includes my name, username, project description, technologies, a code example, a link, and the supplied image. I organized the work into meaningful commits for setup, project changes, selective staging, the About feature, merge, and remote updates.

```
aymennhaila@Aymens-MacBook-Pro git-session1-lurantys % git log --oneline --graph --decorate --all
* 26655a6 (HEAD -> main, origin/main, origin/HEAD) Add source links to assignment report
* aec219d Update assignment report with fetch and pull evidence
* ba30f94 Update README with project notes
* cceea97 Add completed assignment report PDF
* 6d78821 Label the assignment project overview
* 1d72622 Keep important log eligible for tracking
* f92af39 Add About link to README
*   56b395d Merge about page feature
|\
| * 7efa315 (origin/feature/about-page, feature/about-page) Add an about page for the project
|/
* 82bbd81 Update page status and record Git workflow notes
* 02fd3ab Clarify the page title and introduction
* ab321bb Document project and technologies in README
* 8f3c0e3 Ignore local environment and regular logs
* 52087b9 Add starter project files
aymennhaila@Aymens-MacBook-Pro git-session1-lurantys %
```

Current commit history, including the About branch and merge.

## 3. Selective Staging

I changed `index.html` and `app.js` together, staged and committed only `index.html`, and left `app.js` modified. I committed `app.js` later.

```
aymennhaila@Aymens-MacBook-Pro git-session1-lurantys % git status --short
git diff -- index.html app.js

 M app.js
 M index.html
diff --git a/app.js b/app.js
index 39c1371..59fa323 100644
--- a/app.js
+++ b/app.js
@@ -1,5 +1,5 @@
 const statusMessage = document.getElementById("status-message");

 if (statusMessage) {
-  statusMessage.textContent = "Git workflow practice page is ready.";
+  statusMessage.textContent = "The assignment project is ready to explore.";
 }
diff --git a/index.html b/index.html
index 1bb9e54..5be8f76 100644
--- a/index.html
+++ b/index.html
@@ -14,7 +14,7 @@
    <main>
      <section id="overview">
        <h2>Assignment Project Overview</h2>
-       <p>This small project will evolve throughout the assignment.</p>
+       <p>This page demonstrates the Git workflow using the provided starter files..</p>
      </section>
      <section id="status">
        <h2>Status</h2>
aymennhaila@Aymens-MacBook-Pro git-session1-lurantys %
```

Both files changed; only index.html is staged.

```
aymennhaila@Aymens-MacBook-Pro git-session1-lurantys % git add index.html
git status --short
git diff --cached -- index.html
git diff -- app.js

 M app.js
M  index.html
diff --git a/index.html b/index.html
index 1bb9e54..5be8f76 100644
--- a/index.html
+++ b/index.html
@@ -14,7 +14,7 @@
    <main>
      <section id="overview">
        <h2>Assignment Project Overview</h2>
-       <p>This small project will evolve throughout the assignment.</p>
+       <p>This page demonstrates the Git workflow using the provided starter files..</p>
      </section>
      <section id="status">
        <h2>Status</h2>
diff --git a/app.js b/app.js
index 39c1371..59fa323 100644
--- a/app.js
+++ b/app.js
@@ -1,5 +1,5 @@
```

After staging index.html, app.js remains unstaged.

```
 const statusMessage = document.getElementById("status-message");

 if (statusMessage) {
-  statusMessage.textContent = "Git workflow practice page is ready.";
+  statusMessage.textContent = "The assignment project is ready to explore.";
 }
aymennhaila@Aymens-MacBook-Pro git-session1-lurantys %
```

The remaining unstaged diff is app.js.

## 4. Branch & Merge

I created `feature/about-page`, added `ABOUT.md`, and committed the feature on that branch. I switched back to `main` and merged the feature commit; the history shows the merge and the feature branch.

## 5. Remote Changes

A commit records staged changes in local history; a push sends local commits to GitHub. Fetch updates remote-tracking information without changing checked-out files; pull fetches and integrates remote changes. After editing README on GitHub, `git fetch origin` showed local main one commit behind `origin/main`. I then ran `git pull origin main`, which fast-forwarded the README update.

```
aymennhaila@Aymens-MacBook-Pro git-session1-lurantys % git fetch origin
git status -sb
remote: Enumerating objects: 5, done.
remote: Counting objects: 100% (5/5), done.
remote: Compressing objects: 100% (3/3), done.
remote: Total 3 (delta 2), reused 0 (delta 0), pack-reused 0 (from 0)
Unpacking objects: 100% (3/3), 977 bytes | 244.00 KiB/s, done.
From https://github.com/lurantys/git-session1-lurantys
   cceea97..ba30f94  main       -> origin/main
## main...origin/main [behind 1]
 M Assignment-1-Report-Aymen-NHAILA.pdf
aymennhaila@Aymens-MacBook-Pro git-session1-lurantys % git pull origin main

From https://github.com/lurantys/git-session1-lurantys
 * branch            main       -> FETCH_HEAD
Updating cceea97..ba30f94
Fast-forward
 README.md | 2 ++
 1 file changed, 2 insertions(+)
aymennhaila@Aymens-MacBook-Pro git-session1-lurantys %
```

Fetch showed main behind origin/main; pull fast-forwarded the README change.

## 6. Fair Use of AI

I used AI to help draft this report and explain merging branches and resolving conflicts. I asked about ignore rules, selective staging, branch merges, and fetch versus pull. I revised the wording and checked Git behavior with `git status`, `git diff`, `git log`, and `git check-ignore`.

## 7. Sources & Reflection

Sources: git commit, git push, git fetch, git pull, and GitHub Docs: ignoring files.

The hardest concept was understanding conflict resolution. My About branch merged without conflicts; if one arose, I would compare both versions before deciding what to keep. If Git behaved unexpectedly, I would first check the current branch and working tree with `git status -sb`, then inspect the diff.